



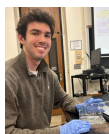
Temperature's effect on turion germination in *Spirodela polyrhiza*

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Introduction

Climate change is increasing global temperatures and has potential to drastically impact ecosystems and the plants within them. To investigate the effects of temperature changes on plant growth, the model organism *Spirodela polyrhiza*, a species of duckweed, was observed. The experiment followed the growth of turions, which are a structure of the duckweed plant that releases from the plant in poor weather conditions to the bottom of their water source. When conditions improve, the turions rise to the surface and germinate, repopulating the duckweed. This study will focus on how our experimental temperatures, 30 °C and 40 °C, compare to the growth of our control temperature, 20 °C



Research Question

How do differing temperatures affect the growth and development of *Spirodela polyrhiza*?

Hypotheses

- 1) Null Hypothesis: Differing temperatures will not have a significant effect on the growth and development of turions for *Spirodela polyrhiza* over time.
- 2) Alternative Hypothesis: Differing temperatures will have a significant effect on the growth and development of turions for *Spirodela polyrhiza* over time.

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Abstract

If germination changes due to change in temperature, the effects of climate change on aquatic plants and their ecosystems can be better examined, and, such, preventative means can be taken to diminish the detrimental effects on water quality and biodiversity. This study sought to determine if temperature has a significant effect on turion germination which was determined by finding percent coverage and number of fronds. It was predicted that as temperature increases, turion germination would increase. ImageJ program was used to find area and Python via Google Collab was used to create 2-way ANOVA and Tukey tests. Temperature had no significant effect on percent coverage and only had a significant impact on one of the three treatments for frond number. Previous studies have shown that many confounding variables also contribute to growth such as light conditions and phosphate concentration in the soil. The results may also prove significant if a greater range of temperature values is used in future studies. This study holds importance as the effects of climate change are spreading considerably and adversely affecting ecosystem survival.

Results

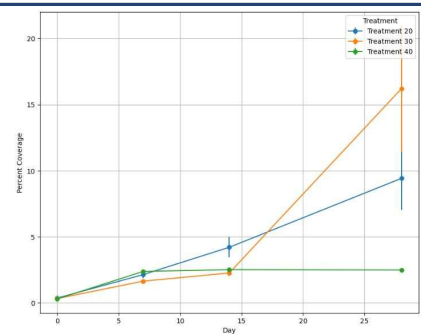


Figure 1. Percent coverage of three different turion treatment groups on Day 0, 7, 14, and 28. Treatment 20 has been kept at 20 degrees Celsius, Treatment 30 has been kept at 30 degrees Celsius, Treatment 40 has been kept at 40 degrees Celsius. Each treatment contains 6 well plates of turions.

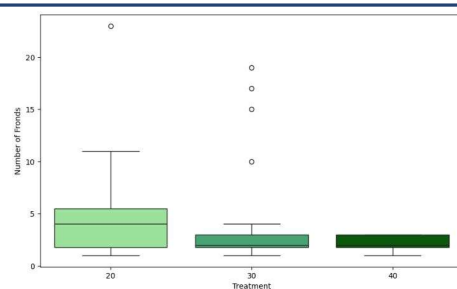


Figure 2. Frond number among three turion treatment groups. Treatment 20 has been kept at 20 degrees Celsius, Treatment 30 has been kept at 30 degrees Celsius, Treatment 40 has been kept at 40 degrees Celsius. Each treatment contains 6 well plates of turions.

Table 4. Tukey Test on the number of fronds among the three turion treatments. Treatment 20 has been kept at 20 degrees Celsius, Treatment 30 has been kept at 30 degrees Celsius, Treatment 40 has been kept at 40 degrees Celsius. Each treatment contains 6 well plates of turions. The results were collected on Day 0, 7, 14, and 28.

group1	group2	meandiff	p-adj	lower	upper	reject
20	30	-0.9167	0.7348	-3.8455	2.0122	False
20	40	-2.9583	0.0472	-5.8872	-0.0295	True
30	40	-2.0417	0.2241	-4.9705	0.8872	False

- Treatments were found to have no significant effect on growth based on the day.
- Percent coverage of both Treatment 20 and 30 increased over the run of the experiment; while, Treatment 40 increased till day 7 and decreased from day 7 to day 28 (Figure 1). None of the differences in percent growth among the treatments was significantly different as none of the t-values were lower than 0.05.

- Treatment 20 produced the greatest range of fronds formed, and Treatments 30 and 40 produced similar ranges (Figure 2).
- Based on Table 4, Treatment 20 resulted in a significantly greater number of fronds formed than treatment 40 (t-value = -0.0295).
- Yet, Treatments 20 and 30 and Treatments 30 and 40 did not result in a significant difference in frond number (Table 4).

Conclusion

The study aimed to determine the effects of varying temperatures on turion frond growth by analyzing percent coverage and frond number over 28 days. While overall growth was observed to be higher in the 20 and 30 degree treatments than in the 40 degree treatment, a significant difference was not found as the p-values were not greater than 0.05. The unexpected results were due to several factors, including inconsistency in data collection and fluctuating temperatures within the building housing the experiment. Should this experiment be replicated, a system is needed in which each task is completed by the same person, as well as ensuring that the temperature of storage is consistent.

Methods

Experimental Design

- 10 mL sterile plant medium
- 1 frond each well
- 3 sets for 3 temperature groups



Data Collection

- ImageJ and excel
- multipoint tool to count fronds
- freehand tool to measure fronds



Data Analysis

- Python Coding Program

Future Directions

Although significance was not found within this experiment, the data highlighted how this experiment can be improved in the future to yield better results. Replicating this experiment could be used to further study the impacts of how climate change and sequential effects on flora will impact an ecosystem's survival. Additionally, this experiment can be expanded to study how different minerals in the water would produce different results when analyzing turion resistance to heat.